

Numerical Physics — Exercise 4

Electrostatics: cylinder with variable dielectric and free charges. Shooting method, finite differences, finite elements.

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1 Introduction

In this exercise, we study an electrostatic problem in cylindrical geometry with radial symmetry, in the presence of free charges and of a dielectric medium whose relative permittivity may vary with the radius. The objective is to determine the electrostatic potential $\phi(r)$, the corresponding radial electric field $E_r(r)$, and the electric displacement field $D_r(r)$, under appropriate boundary conditions at the outer radius.

The problem is particularly well suited to the comparison of several standard numerical approaches used in physics. On the one hand, its cylindrical symmetry allows the governing equations to be reduced to a one-dimensional radial problem, which makes the analytical derivations tractable in simple benchmark cases. On the other hand, the presence of non-uniform material properties and piecewise-defined charge distributions provides a natural framework for testing more general numerical methods.

The work is divided into two complementary parts. First, we derive the electrostatic equation satisfied by the potential and establish several analytical results that serve as reference solutions. These include the uniform case, a non-trivial case for which an exact expression can still be obtained, the reformulation of the problem for a shooting method, the finite-difference discretization, and the variational formulation underlying the finite-element method. Second, we implement and compare the corresponding numerical methods: a shooting method combined with a fourth-order Runge–Kutta scheme, a finite-difference discretization leading to a tridiagonal linear system, and a finite-element formulation adapted to heterogeneous dielectric profiles.

The numerical study is then used to validate the implementations and to analyse more general electrostatic configurations. In particular, we compare numerical and analytical solutions in benchmark cases, study convergence under mesh refinement, investigate a non-trivial dielectric and free-charge distribution, verify numerically the relation $\nabla \cdot \mathbf{D} = \rho_{\text{lib}}$, and finally explore the effect of modifying the dielectric and charge profiles.

This report is organised as follows. Section 2 presents the analytical development of the problem. Section 3 summarises the numerical methods and implementation choices. Section 4 gathers the simulations and the discussion of the numerical results. The final sections are devoted to the conclusion and references.

2 Analytical development

2.1 a) Differential equation

Given the electrostatic equations in matter:

$$\nabla \cdot \mathbf{D} = \rho_{\text{lib}} \tag{1}$$

$$\mathbf{D} = \varepsilon_0 \varepsilon_r(\mathbf{x}) \mathbf{E} \tag{2}$$

$$\mathbf{E} = -\nabla \phi \tag{3}$$

We start by substituting (3) into the displacement field (2)

$$\mathbf{D} = -\varepsilon_0 \varepsilon_r(\mathbf{x}) \nabla \phi.$$

Using Gauss' law (1), we obtain the following.

$$\nabla \cdot (-\varepsilon_0 \varepsilon_r(\mathbf{x}) \nabla \phi) = \rho_{\text{lib}}.$$

Since ε_0 is constant, the general differential equation for the electrostatic potential in an inhomogeneous dielectric medium is

$$\boxed{\nabla \cdot (\varepsilon_r(\mathbf{x}) \nabla \phi) = -\frac{\rho_{\text{lib}}(\mathbf{x})}{\varepsilon_0}}. \quad (4)$$

In the problem considered here, the system is cylindrically symmetric and sufficiently long in the z direction, so edge effects can be neglected. That is why all physical quantities depend only on the radial coordinate r .

In cylindrical coordinates, for a purely radial vector field,

$$\nabla \cdot (A_r(r) \mathbf{e}_r) = \frac{1}{r} \frac{d}{dr} (r A_r(r)). \quad (5)$$

Here,

$$\varepsilon_r(r) \nabla \phi = \varepsilon_r(r) \frac{d\phi}{dr} \mathbf{e}_r.$$

Thus, the differential equation becomes

$$\boxed{\frac{1}{r} \frac{d}{dr} \left(r \varepsilon_r(r) \frac{d\phi}{dr} \right) = -\frac{\rho_{\text{lib}}(r)}{\varepsilon_0}}. \quad (6)$$

Or equivalently,

$$\frac{d}{dr} \left(r \varepsilon_r(r) \frac{d\phi}{dr} \right) = -r \frac{\rho_{\text{lib}}(r)}{\varepsilon_0}$$

The boundary condition imposed at the outer radius is

$$\phi(R) = V_0.$$

In the present exercise, unless otherwise stated, $V_0 = 0$.

2.2 b) Analytical solution for the uniform case

We first consider the uniform test case

$$\varepsilon_r(r) = 1, \quad \rho_{\text{lib}}(r) = \varepsilon_0, \quad V_0 = 0 \quad (7)$$

Therefore,

$$\frac{\rho_{\text{lib}}}{\varepsilon_0} = 1$$

The radial differential equation then becomes

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{d\phi}{dr} \right) = -1.$$

Multiplying by r , we get

$$\frac{d}{dr} \left(r \frac{d\phi}{dr} \right) = -r.$$

Integrating once with respect to r gives

$$r \frac{d\phi}{dr} = -\frac{r^2}{2} + C_1.$$

Dividing by r ,

$$\frac{d\phi}{dr} = -\frac{r}{2} + \frac{C_1}{r}.$$

The electrostatic potential and the electric field must remain finite at the origin ($r=0$). Which implies that the constant:

$$C_1 = 0.$$

Thus, solving the following equation,

$$\frac{d\phi}{dr} = -\frac{r}{2} \quad (8)$$

Using

$$E_r(r) = -\frac{d\phi}{dr}, \quad (9)$$

we obtain

$$\boxed{E_r(r) = \frac{r}{2}}. \quad (10)$$

Solving again (8) for ϕ we obtain

$$\phi(r) = -\frac{r^2}{4} + C_2.$$

The boundary condition is

$$\phi(R) = 0. \quad (11)$$

Therefore,

$$C_2 = \frac{R^2}{4}.$$

And the final analytical solution is:

$$\boxed{\phi(r) = \frac{R^2 - r^2}{4}} \quad (12)$$

This result is physically consistent. By cylindrical symmetry, the electric field must vanish at the origin. In addition, since the enclosed charge inside a cylinder of radius r grows as r^2 , the radial electric field grows linearly with r .

2.3 c) Analytical solution for the case $b = R$

We now consider the case

$$\varepsilon_r(r) = 1, \quad \rho_{\text{lib}}(r) = \varepsilon_0 a_0 \sin\left(\frac{\pi r}{R}\right), \quad V_0 = 0 \quad (13)$$

We then have

$$\frac{\rho_{\text{lib}}(r)}{\varepsilon_0} = a_0 \sin\left(\frac{\pi r}{R}\right)$$

And the equation for the potential is

$$\frac{d}{dr} \left(r \frac{d\phi}{dr} \right) = -a_0 r \sin\left(\frac{\pi r}{R}\right).$$

Let

$$k = \frac{\pi}{R}. \quad (14)$$

Then the equation becomes

$$\frac{d}{dr} \left(r \frac{d\phi}{dr} \right) = -a_0 r \sin(kr).$$

Integrating from 0 to r ,

$$r \frac{d\phi}{dr} = -a_0 \int_0^r s \sin(ks) ds.$$

The integration constant has been set to zero to ensure regularity at $r = 0$. Otherwise, after dividing by r , a singular term proportional to $1/r$ would appear.

We need to compute the integral

$$\int s \sin(ks) ds. \quad (15)$$

Using integration by parts,

$$\int s \sin(ks) ds = -\frac{s \cos(ks)}{k} + \frac{\sin(ks)}{k^2}.$$

Substituting this result,

$$\begin{aligned} r \frac{d\phi}{dr} &= -a_0 \left[-\frac{r \cos(kr)}{k} + \frac{\sin(kr)}{k^2} \right] \\ \frac{d\phi}{dr} &= a_0 \left[\frac{\cos(kr)}{k} - \frac{\sin(kr)}{rk^2} \right]. \end{aligned} \quad (16)$$

Therefore, since

$$\begin{aligned} E_r(r) &= -\frac{d\phi}{dr}, \\ E_r(r) &= a_0 \left[\frac{\sin(kr)}{rk^2} - \frac{\cos(kr)}{k} \right]. \end{aligned} \quad (17)$$

Substituting back $k = \pi/R$, we obtain

$$E_r(r) = a_0 \left[\frac{R^2}{\pi^2 r} \sin\left(\frac{\pi r}{R}\right) - \frac{R}{\pi} \cos\left(\frac{\pi r}{R}\right) \right]. \quad (18)$$

Although this expression contains a factor $1/r$, the field remains finite at the origin because

$$\sin\left(\frac{\pi r}{R}\right) \sim \frac{\pi r}{R} \quad \text{as } r \rightarrow 0.$$

In fact, one finds $E_r(0) = 0$ as required by cylindrical symmetry.

We now derive the potential. From the previous expression (16), we integrate to find the solution for the potential.

$$\phi(r) = a_0 \left[\int \frac{\cos(kr)}{k} dr - \int \frac{\sin(kr)}{rk^2} dr \right] + C_2. \quad (19)$$

The first integral is

$$\int \frac{\cos(kr)}{k} dr = \frac{\sin(kr)}{k^2}. \quad (20)$$

For the second integral, we introduce the sine integral function as a variable,

$$\text{Si}(x) = \int_0^x \frac{\sin u}{u} du. \quad (21)$$

Then,

$$\int \frac{\sin(kr)}{r} dr = \text{Si}(kr). \quad (22)$$

Therefore putting all these results together,

$$\phi(r) = \frac{a_0}{k^2} [\sin(kr) - \text{Si}(kr)] + C_2. \quad (23)$$

We now impose the boundary condition

$$\phi(R) = 0. \quad (24)$$

Since $kR = \pi$, we have

$$\sin(kR) = \sin(\pi) = 0.$$

Thus,

$$0 = \frac{a_0}{k^2} [-\text{Si}(\pi)] + C_2.$$

And,

$$C_2 = \frac{a_0}{k^2} \text{Si}(\pi).$$

The analytical solution for the potential is therefore

$$\boxed{\phi(r) = \frac{a_0 R^2}{\pi^2} \left[\text{Si}(\pi) + \sin\left(\frac{\pi r}{R}\right) - \text{Si}\left(\frac{\pi r}{R}\right) \right]}. \quad (25)$$

And the corresponding electric field

$$\boxed{E_r(r) = a_0 \left[\frac{R^2}{\pi^2 r} \sin\left(\frac{\pi r}{R}\right) - \frac{R}{\pi} \cos\left(\frac{\pi r}{R}\right) \right]}. \quad (26)$$

At the outer boundary, $r = R$, this gives

$$E_r(R) = a_0 \left[\frac{R^2}{\pi^2 R} \sin(\pi) - \frac{R}{\pi} \cos(\pi) \right].$$

Using

$$\sin(\pi) = 0, \quad \cos(\pi) = -1,$$

we obtain

$$\boxed{E_r(R) = \frac{a_0 R}{\pi}}. \quad (27)$$

This value will be useful later as a reference when comparing the analytical result with the numerical shooting method.

2.4 d) Shooting method

We now rewrite the uniform problem as an initial value problem suitable for a shooting method. In the uniform case,

$$\varepsilon_r(r) = 1, \quad \frac{\rho_{\text{lib}}}{\varepsilon_0} = 1.$$

The radial equation is

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{d\phi}{dr} \right) = -1.$$

This can be written as

$$\frac{1}{r} \frac{d}{dr} (r E_r) = 1.$$

Expanding the derivative gives

$$\frac{dE_r}{dr} + \frac{1}{r} E_r = 1. \quad (28)$$

The exercise asks us to use the vector $\begin{pmatrix} \phi \\ -E_r \end{pmatrix}$. We therefore define:

$$y_1(r) = \phi(r), \quad y_2(r) = -E_r(r). \quad (29)$$

We have that

$$y_2(r) = \frac{d\phi}{dr}. \quad (30)$$

Therefore,

$$\frac{dy_1}{dr} = y_2. \quad (31)$$

The equation for E_r is

$$\frac{dE_r}{dr} = 1 - \frac{E_r}{r}.$$

Since $y_2 = -E_r$, we get

$$\frac{dy_2}{dr} = -1 - \frac{y_2}{r}.$$

Therefore, in terms of r , the first-order system is

$$\boxed{\frac{d}{dr} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} y_2 \\ -1 - \frac{y_2}{r} \end{pmatrix}}. \quad (32)$$

The shooting method is performed using the change of variable

$$x = 1 - \frac{r}{R}. \quad (33)$$

Hence, $r = R(1 - x)$, and $\frac{dr}{dx} = -R$. Using the chain rule,

$$\frac{d}{dx} = \frac{dr}{dx} \frac{d}{dr} = -R \frac{d}{dr}.$$

Therefore,

$$\frac{dy_1}{dx} = -R y_2, \quad (34)$$

and

$$\frac{dy_2}{dx} = -R \left(-1 - \frac{y_2}{r} \right). \quad (35)$$

Since $r = R(1 - x)$, this gives

$$\frac{dy_2}{dx} = R + \frac{R y_2}{R(1 - x)}.$$

Thus,

$$\frac{dy_2}{dx} = R + \frac{y_2}{1 - x}. \quad (36)$$

The initial value problem in the variable x is therefore

$$\boxed{\frac{d}{dx} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} -R y_2 \\ R + \frac{y_2}{1 - x} \end{pmatrix}}. \quad (37)$$

The boundary condition at the outer radius is

$$\phi(R) = V_0.$$

Since $x = 0$ corresponds to $r = R$, this becomes the initial condition

$$y_1(0) = V_0.$$

The value of the electric field at the boundary is not known a priori. We therefore introduce a shooting parameter

$$E_R = E_r(R). \quad (38)$$

Since

$$y_2 = -E_r,$$

we have

$$y_2(0) = -E_R. \quad (39)$$

Thus, the initial conditions for the shooting method are

$$\boxed{y_1(0) = V_0, \quad y_2(0) = -E_R}. \quad (40)$$

The shooting parameter E_R must be chosen such that the inner boundary condition at $r = 0$ is satisfied. By cylindrical symmetry, the electric field must vanish at the origin:

$$E_r(0) = 0.$$

Since $x = 1$ corresponds to $r = 0$, this final constraint becomes

$$\boxed{y_2(1) = -E_r(0) = 0}. \quad (41)$$

The shooting method therefore consists of varying the initial value E_R until the solution of the initial value problem satisfies

$$y_2(1) = 0.$$

2.5 e) Finite differences method

We now construct a finite difference discretization of the radial equation. Starting from

$$\frac{1}{r} \frac{d}{dr} \left(r \varepsilon_r(r) \frac{d\phi}{dr} \right) = -\frac{\rho_{\text{lib}}(r)}{\varepsilon_0},$$

we first multiply by r in order to avoid the apparent singularity at $r = 0$:

$$\frac{d}{dr} \left(r \varepsilon_r(r) \frac{d\phi}{dr} \right) = -r \frac{\rho_{\text{lib}}(r)}{\varepsilon_0}.$$

We introduce a uniform grid with spacing

$$h = \frac{R}{N}. \quad (42)$$

Following the statement of the problem, we place the unknowns on the half-grid:

$$r_{k+1/2} = \left(k + \frac{1}{2} \right) h, \quad k = 0, \dots, N-1. \quad (43)$$

We denote

$$\phi_k \approx \phi(r_{k+1/2}).$$

For an interior interface, we approximate

$$\left. \frac{d\phi}{dr} \right|_{r_k} \approx \frac{\phi_k - \phi_{k-1}}{h}. \quad (44)$$

We define

$$\varepsilon_k = \varepsilon_r(r_k), \quad q_k = \frac{\rho_{\text{lib}}(r_{k+1/2})}{\varepsilon_0}.$$

Then, at the half-grid point $r_{k+1/2}$, the finite difference equation is

$$\frac{1}{h} \left[r_{k+1} \varepsilon_{k+1} \frac{\phi_{k+1} - \phi_k}{h} - r_k \varepsilon_k \frac{\phi_k - \phi_{k-1}}{h} \right] = -r_{k+1/2} q_k.$$

Multiplying by $-h^2$, we obtain

$$-r_k \varepsilon_k \phi_{k-1} + (r_k \varepsilon_k + r_{k+1} \varepsilon_{k+1}) \phi_k - r_{k+1} \varepsilon_{k+1} \phi_{k+1} = h^2 r_{k+1/2} q_k.$$

This gives a system for the unknowns

$$\phi_0, \phi_1, \dots, \phi_{N-1}.$$

At the inner boundary, $r = 0$, no explicit boundary condition has to be imposed. Indeed, the first interface is located at

$$r_0 = 0.$$

Therefore, the corresponding flux term vanishes automatically:

$$r_0 \varepsilon_0 \frac{\phi_0 - \phi_{-1}}{h} = 0.$$

This implicitly enforces the regularity condition at the origin, equivalent to

$$E_r(0) = 0.$$

At the outer boundary, we impose the Dirichlet condition

$$\phi(R) = V_0.$$

We introduce a ghost value ϕ_N at

$$r_{N+1/2} = R + \frac{h}{2}.$$

Using a centered interpolation at $r = R$,

$$\phi(R) \approx \frac{\phi_{N-1} + \phi_N}{2} = V_0.$$

Therefore,

$$\phi_N = 2V_0 - \phi_{N-1}.$$

For the last equation, corresponding to $k = N - 1$, we substitute this ghost value. The resulting last row becomes

$$-r_{N-1} \varepsilon_{N-1} \phi_{N-2} + (r_{N-1} \varepsilon_{N-1} + 2r_N \varepsilon_N) \phi_{N-1} = h^2 r_{N-1/2} q_{N-1} + 2r_N \varepsilon_N V_0. \quad (45)$$

Thus, the finite difference method leads to a tridiagonal linear system of the form

$$\boxed{A\Phi = b} \quad (46)$$

where

$$\Phi = \begin{pmatrix} \phi_0 \\ \phi_1 \\ \vdots \\ \phi_{N-1} \end{pmatrix}.$$

For interior rows, the coefficients are

$$A_{k,k-1} = -r_k \varepsilon_k, \quad (47)$$

$$A_{k,k} = r_k \varepsilon_k + r_{k+1} \varepsilon_{k+1}, \quad (48)$$

$$A_{k,k+1} = -r_{k+1} \varepsilon_{k+1}, \quad (49)$$

and

$$b_k = h^2 r_{k+1/2} q_k. \quad (50)$$

The first row automatically satisfies the regularity condition at the origin because $r_0 = 0$. The last row is modified to impose the boundary condition $\phi(R) = V_0$ by centered interpolation.

2.6 f) Application to the uniform case

We now solve the finite difference system by hand for the uniform case discussed in part (b). We take

$$\varepsilon_r = 1, \quad \frac{\rho_{\text{lib}}}{\varepsilon_0} = 1, \quad V_0 = 0.$$

We use

$$N = 4, \quad h = \frac{R}{4}.$$

Thus the half-grid points are

$$r_{1/2} = \frac{R}{8}, \quad r_{3/2} = \frac{3R}{8}, \quad r_{5/2} = \frac{5R}{8}, \quad r_{7/2} = \frac{7R}{8}.$$

The unknowns are

$$\Phi = \begin{pmatrix} \phi_0 \\ \phi_1 \\ \phi_2 \\ \phi_3 \end{pmatrix},$$

where

$$\phi_k \approx \phi(r_{k+1/2}). \quad (51)$$

For the uniform case, after dividing each equation by h , we obtain (46) with

$$A = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 3 & -2 & 0 \\ 0 & -2 & 5 & -3 \\ 0 & 0 & -3 & 11 \end{pmatrix},$$

and

$$b = R^2 \begin{pmatrix} \frac{1}{32} \\ \frac{3}{32} \\ \frac{5}{32} \\ \frac{7}{32} \end{pmatrix}.$$

The last diagonal coefficient is 11 rather than 7 because the Dirichlet boundary condition at $r = R$ is imposed by the centered interpolation

$$\phi_4 = 2V_0 - \phi_3.$$

Since here $V_0 = 0$, this gives

$$\phi_4 = -\phi_3. \quad (52)$$

We now perform Gaussian elimination. The first pivot is 1. The multiplier used to eliminate the coefficient below it is

$$m_{21} = -1.$$

After the first elimination step, the second row of A becomes

$$(0, 2, -2, 0).$$

The second pivot is 2. The multiplier used to eliminate the coefficient below it is

$$m_{32} = -1.$$

After the second elimination step, the third row becomes

$$(0, 0, 3, -3).$$

The third pivot is 3. The multiplier used to eliminate the coefficient below it is

$$m_{43} = -1.$$

After the third elimination step, the last row becomes

$$(0, 0, 0, 8).$$

We now solve the system by Gaussian elimination. The augmented system is

$$\left[\begin{array}{cccc|c} 1 & -1 & 0 & 0 & \frac{R^2}{32} \\ -1 & 3 & -2 & 0 & \frac{3R^2}{32} \\ 0 & -2 & 5 & -3 & \frac{5R^2}{32} \\ 0 & 0 & -3 & 11 & \frac{7R^2}{32} \end{array} \right].$$

We perform

$$R_2 \leftarrow R_2 - m_{21}R_1 = R_2 + R_1.$$

This gives

$$\left[\begin{array}{cccc|c} 1 & -1 & 0 & 0 & \frac{R^2}{32} \\ 0 & 2 & -2 & 0 & \frac{4R^2}{32} \\ 0 & -2 & 5 & -3 & \frac{5R^2}{32} \\ 0 & 0 & -3 & 11 & \frac{7R^2}{32} \end{array} \right].$$

We perform

$$R_3 \leftarrow R_3 - m_{32}R_2 = R_3 + R_2.$$

Thus,

$$\left[\begin{array}{cccc|c} 1 & -1 & 0 & 0 & \frac{R^2}{32} \\ 0 & 2 & -2 & 0 & \frac{4R^2}{32} \\ 0 & 0 & 3 & -3 & \frac{9R^2}{32} \\ 0 & 0 & -3 & 11 & \frac{7R^2}{32} \end{array} \right].$$

We perform

$$R_4 \leftarrow R_4 - m_{43}R_3 = R_4 + R_3.$$

This gives

$$\left[\begin{array}{cccc|c} 1 & -1 & 0 & 0 & \frac{R^2}{32} \\ 0 & 2 & -2 & 0 & \frac{4R^2}{32} \\ 0 & 0 & 3 & -3 & \frac{9R^2}{32} \\ 0 & 0 & 0 & 8 & \frac{16R^2}{32} \end{array} \right].$$

We can now solve by back substitution. From the last row,

$$8\phi_3 = \frac{R^2}{2},$$

so

$$\phi_3 = \frac{R^2}{16}.$$

From the third row,

$$3\phi_2 - 3\phi_3 = \frac{9R^2}{32}.$$

Substituting $\phi_3 = R^2/16$,

$$3\phi_2 = \frac{9R^2}{32} + 3\frac{R^2}{16} = \frac{9R^2}{32} + \frac{6R^2}{32} = \frac{15R^2}{32}.$$

Thus,

$$\phi_2 = \frac{5R^2}{32}.$$

From the second row,

$$2\phi_1 - 2\phi_2 = \frac{R^2}{8}.$$

Substituting $\phi_2 = 5R^2/32$,

$$2\phi_1 = \frac{R^2}{8} + 2\frac{5R^2}{32} = \frac{4R^2}{32} + \frac{10R^2}{32} = \frac{14R^2}{32}.$$

Therefore,

$$\phi_1 = \frac{7R^2}{32}.$$

Finally, from the first row,

$$\phi_0 - \phi_1 = \frac{R^2}{32}.$$

Substituting $\phi_1 = 7R^2/32$,

$$\phi_0 = \frac{R^2}{32} + \frac{7R^2}{32} = \frac{8R^2}{32} = \frac{R^2}{4}.$$

Hence, the solution obtained by Gaussian elimination is

$$\Phi = \begin{pmatrix} \frac{R^2}{4} \\ \frac{7R^2}{32} \\ \frac{5R^2}{32} \\ \frac{R^2}{16} \end{pmatrix}. \quad (53)$$

On the other hand, the matrices L and U are

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{pmatrix}, \quad (54)$$

and

$$U = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 2 & -2 & 0 \\ 0 & 0 & 3 & -3 \\ 0 & 0 & 0 & 8 \end{pmatrix}. \quad (55)$$

Thus,

$$A = LU. \quad (56)$$

We first solve

$$Lz = b. \quad (57)$$

This gives

$$\begin{aligned} z_1 &= \frac{R^2}{32}, \\ z_2 &= \frac{R^2}{8}, \\ z_3 &= \frac{9R^2}{32}, \end{aligned}$$

$$z_4 = \frac{R^2}{2}.$$

Then we solve

$$U\Phi = z. \quad (58)$$

Back substitution gives

$$\begin{aligned} 8\phi_3 &= \frac{R^2}{2}, \\ \phi_3 &= \frac{R^2}{16}. \end{aligned}$$

Then,

$$3\phi_2 - 3\phi_3 = \frac{9R^2}{32},$$

so

$$\phi_2 = \frac{5R^2}{32}.$$

Next we have,

$$2\phi_1 - 2\phi_2 = \frac{R^2}{8},$$

so

$$\phi_1 = \frac{7R^2}{32}.$$

Finally,

$$\phi_0 - \phi_1 = \frac{R^2}{32},$$

so

$$\phi_0 = \frac{R^2}{4}.$$

Therefore, the numerical solution is

$$\Phi = \begin{pmatrix} \frac{R^2}{4} \\ \frac{7R^2}{32} \\ \frac{5R^2}{32} \\ \frac{R^2}{16} \end{pmatrix}. \quad (59)$$

This is the same result as the one obtained by direct Gaussian elimination, confirming the equivalence between Gaussian elimination and the LU decomposition.

We now compare with the exact solution from part (b),

$$\phi_{\text{exact}}(r) = \frac{R^2 - r^2}{4}. \quad (60)$$

The numerical unknowns are located at

$$r_{k+1/2} = \left(k + \frac{1}{2}\right) \frac{R}{4}. \quad (61)$$

Thus, the exact values are

$$\phi_{\text{exact}}\left(\frac{R}{8}\right) = \frac{63R^2}{256}, \quad \phi_{\text{exact}}\left(\frac{3R}{8}\right) = \frac{55R^2}{256}, \quad \phi_{\text{exact}}\left(\frac{5R}{8}\right) = \frac{39R^2}{256}, \quad \phi_{\text{exact}}\left(\frac{7R}{8}\right) = \frac{15R^2}{256}$$

The numerical values can be written as

$$\phi_0 = \frac{64R^2}{256}, \quad \phi_1 = \frac{56R^2}{256}, \quad \phi_2 = \frac{40R^2}{256}, \quad \phi_3 = \frac{16R^2}{256}$$

Therefore, the numerical solution differs from the exact solution by

$$\phi_k - \phi_{\text{exact}}(r_{k+1/2}) = \frac{R^2}{256}, \quad k = 0, 1, 2, 3. \quad (62)$$

The finite difference solution is therefore very close to the exact solution, even for the very coarse grid $N = 4$. The error is uniform on the half-grid points and is of order h^2 , as expected for a centered finite difference discretization.

Finally, we reuse the same matrices L and U to solve the problem without free charge but with a non-zero boundary potential,

$$\rho_{\text{lib}} = 0, \quad V_0 = 1 \text{ V}.$$

In this case, the right-hand side only receives a contribution from the boundary condition at $r = R$. Since $N = 4$, the discrete system is

$$A\Phi = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 8 \end{pmatrix}.$$

Solving this system using the same LU factorization gives

$$\boxed{\Phi = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}}. \quad (63)$$

This agrees exactly with the analytical solution. Indeed, when there is no free charge, the equation is

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{d\phi}{dr} \right) = 0.$$

Integrating,

$$r \frac{d\phi}{dr} = C_1.$$

Regularity at $r = 0$ requires

$$C_1 = 0.$$

Therefore,

$$\frac{d\phi}{dr} = 0,$$

so the potential is constant:

$$\phi(r) = C_2.$$

Using the boundary condition

$$\phi(R) = 1, \quad \boxed{\phi(r) = 1}.$$

Hence, the finite difference solution reproduces the exact solution exactly in the charge-free case.

2.7 g) Variational formulation in general geometry

We now derive the variational formulation of the electrostatic problem in a general domain Ω . The strong form of the equation is

$$\nabla \cdot (\varepsilon_r(\mathbf{x}) \nabla \phi) = -\frac{\rho_{\text{lib}}(\mathbf{x})}{\varepsilon_0} \quad \text{in } \Omega, \quad (64)$$

with the Dirichlet boundary condition

$$\phi = V_0 \quad \text{on } \partial\Omega.$$

To derive the weak form, we introduce a test function $\eta(\mathbf{x})$ such that

$$\eta = 0 \quad \text{on } \partial\Omega.$$

We multiply the differential equation by η and integrate over the domain:

$$\int_{\Omega} \eta \nabla \cdot (\varepsilon_r \nabla \phi) d\Omega = - \int_{\Omega} \eta \frac{\rho_{\text{lib}}}{\varepsilon_0} d\Omega.$$

We now integrate by parts the left-hand side. Using Green's identity,

$$\int_{\Omega} \eta \nabla \cdot (\varepsilon_r \nabla \phi) d\Omega = \int_{\partial\Omega} \eta \varepsilon_r \nabla \phi \cdot \mathbf{n} dS - \int_{\Omega} \varepsilon_r \nabla \eta \cdot \nabla \phi d\Omega. \quad (65)$$

Since the test function vanishes on the boundary, the boundary term is zero:

$$\int_{\partial\Omega} \eta \varepsilon_r \nabla \phi \cdot \mathbf{n} dS = 0. \quad (66)$$

Therefore, the weak form becomes

$$\boxed{\int_{\Omega} \varepsilon_r \nabla \eta \cdot \nabla \phi d\Omega = \int_{\Omega} \eta \frac{\rho_{\text{lib}}}{\varepsilon_0} d\Omega}. \quad (67)$$

This is the variational formulation of the electrostatic problem. The unknown potential ϕ satisfies the prescribed Dirichlet boundary condition on $\partial\Omega$, while the test functions vanish on the boundary.

2.8 h) Variational formulation in cylindrical symmetry

We now specialize the variational formulation to the cylindrical geometry of the problem. The cylinder is assumed to be sufficiently long, and all quantities are independent of the angular coordinate θ and of the axial coordinate z . The general weak form is

$$\int_{\Omega} \varepsilon_r \nabla \eta \cdot \nabla \phi d\Omega = \int_{\Omega} \eta \frac{\rho_{\text{lib}}}{\varepsilon_0} d\Omega. \quad (68)$$

In cylindrical coordinates,

$$d\Omega = r dr d\theta dz. \quad (69)$$

Since ϕ and η depend only on r , their gradients are purely radial:

$$\nabla \phi = \frac{d\phi}{dr} \mathbf{e}_r, \quad \nabla \eta = \frac{d\eta}{dr} \mathbf{e}_r. \quad (70)$$

Therefore,

$$\nabla \eta \cdot \nabla \phi = \frac{d\eta}{dr} \frac{d\phi}{dr}.$$

The left-hand side becomes

$$\int_0^{L_z} \int_0^{2\pi} \int_0^R \varepsilon_r(r) \frac{d\eta}{dr} \frac{d\phi}{dr} r dr d\theta dz.$$

Since the integrand is independent of θ and z , the integrations over those variables give a multiplicative factor:

$$\int_0^{L_z} dz \int_0^{2\pi} d\theta = 2\pi L_z.$$

Thus, the left-hand side is

$$2\pi L_z \int_0^R \varepsilon_r(r) \frac{d\eta}{dr} \frac{d\phi}{dr} r dr.$$

Similarly, the right-hand side becomes

$$2\pi L_z \int_0^R \eta(r) \frac{\rho_{\text{lib}}(r)}{\varepsilon_0} r dr.$$

The common factor $2\pi L_z$ cancels from both sides. Therefore, the one-dimensional variational formulation is

$$\boxed{\int_0^R \varepsilon_r(r) \frac{d\eta}{dr} \frac{d\phi}{dr} r dr = \int_0^R \eta(r) \frac{\rho_{\text{lib}}(r)}{\varepsilon_0} r dr.} \quad (71)$$

2.9 i) Matrix elements and right-hand side

We discretize the one-dimensional weak formulation obtained in the previous section:

$$\int_0^R \varepsilon_r(r) \frac{d\eta}{dr} \frac{d\phi}{dr} r dr = \int_0^R \eta(r) \frac{\rho_{\text{lib}}(r)}{\varepsilon_0} r dr.$$

The interval $[0, R]$ is split into two parts. The first part, $[0, b]$, is divided into N_1 equidistant intervals of size

$$h_1 = \frac{b}{N_1},$$

and the second part, $[b, R]$, is divided into N_2 equidistant intervals of size

$$h_2 = \frac{R - b}{N_2}.$$

Thus, the mesh nodes are

$$r_k = kh_1, \quad k = 0, \dots, N_1,$$

in the first region, and

$$r_{N_1+j} = b + jh_2, \quad j = 0, \dots, N_2,$$

in the second region.

For a generic element

$$I_k = [r_k, r_{k+1}],$$

we define

$$h_k = r_{k+1} - r_k, \quad r_{k+1/2} = \frac{r_k + r_{k+1}}{2}.$$

The approximate finite element solution is written as

$$\phi_h(r) = \sum_j \Phi_j N_j(r),$$

where N_j are the usual piecewise linear basis functions.

On the element $[r_k, r_{k+1}]$, the two local basis functions are

$$N_k(r) = \frac{r_{k+1} - r}{h_k}, \quad N_{k+1}(r) = \frac{r - r_k}{h_k}.$$

Their derivatives are constant:

$$\frac{dN_k}{dr} = -\frac{1}{h_k}, \quad \frac{dN_{k+1}}{dr} = \frac{1}{h_k}.$$

The local stiffness matrix is defined by

$$K_{ij}^{(k)} = \int_{r_k}^{r_{k+1}} \varepsilon_r(r) \frac{dN_i}{dr} \frac{dN_j}{dr} r \, dr, \quad i, j \in \{k, k+1\}.$$

Using the midpoint rule on the element,

$$\varepsilon_r(r)r \approx \varepsilon_r(r_{k+1/2})r_{k+1/2}.$$

Therefore,

$$K_{ij}^{(k)} \approx \varepsilon_r(r_{k+1/2})r_{k+1/2} \frac{dN_i}{dr} \frac{dN_j}{dr} h_k.$$

Since the derivatives are $\pm 1/h_k$, the local stiffness matrix is

$$\boxed{K^{(k)} = \alpha_k \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}} \quad (72)$$

with

$$\boxed{\alpha_k = \frac{\varepsilon_r(r_{k+1/2})r_{k+1/2}}{h_k}}. \quad (73)$$

The local right-hand side vector is defined by

$$f_i^{(k)} = \int_{r_k}^{r_{k+1}} N_i(r) \frac{\rho_{\text{lib}}(r)}{\varepsilon_0} r \, dr.$$

Again, using the midpoint rule,

$$\frac{\rho_{\text{lib}}(r)}{\varepsilon_0} r \approx \frac{\rho_{\text{lib}}(r_{k+1/2})}{\varepsilon_0} r_{k+1/2}.$$

At the midpoint of the element,

$$N_k(r_{k+1/2}) = \frac{1}{2}, \quad N_{k+1}(r_{k+1/2}) = \frac{1}{2}.$$

Therefore,

$$f_k^{(k)} \approx \frac{h_k}{2} \frac{\rho_{\text{lib}}(r_{k+1/2})}{\varepsilon_0} r_{k+1/2},$$

and

$$f_{k+1}^{(k)} \approx \frac{h_k}{2} \frac{\rho_{\text{lib}}(r_{k+1/2})}{\varepsilon_0} r_{k+1/2}.$$

Hence, the local load vector is

$$\boxed{f^{(k)} = \beta_k \begin{pmatrix} 1 \\ 1 \end{pmatrix}} \quad (74)$$

where

$$\beta_k = \frac{h_k}{2} \frac{\rho_{\text{lib}}(r_{k+1/2})}{\varepsilon_0} r_{k+1/2}. \quad (75)$$

Each element contributes to the global matrix as

$$\begin{aligned} A_{k,k} &+= \alpha_k, & A_{k,k+1} &+= -\alpha_k, \\ A_{k+1,k} &+= -\alpha_k, & A_{k+1,k+1} &+= \alpha_k, \end{aligned}$$

and to the global right-hand side as

$$b_k += \beta_k, \quad b_{k+1} += \beta_k.$$

After assembling all the intervals, the global finite element system has the form

$$A\Phi = b. \quad (76)$$

The Dirichlet boundary condition at the outer radius is

$$\phi(R) = V_0.$$

Since $r = R$ corresponds to the last node, this condition is imposed as

$$\Phi_{N_1+N_2} = V_0.$$

No essential boundary condition is imposed at $r = 0$. The regularity condition at the origin is naturally satisfied by the weak formulation because of the cylindrical weight r .

3 Numerical method and implementation

3.1 Shooting method

The shooting method transforms the original boundary value problem into a family of initial value problems. The unknown value of the electric field at the outer boundary, $E_r(R)$, is treated as a shooting parameter. For each trial value, the first-order system derived in Section 2.4 is integrated numerically from the outer boundary toward the origin. The correct shooting parameter is the one for which the regularity condition $E_r(0) = 0$ is satisfied.

In practice, the first-order system is solved with a fourth-order Runge–Kutta scheme in the variable

$$x = 1 - \frac{r}{R},$$

so that the integration interval is $x \in [0, 1]$. The shooting residual is defined from the numerical value of the electric field near the origin, and a secant method is used to update the trial values of $E_r(R)$ until the residual becomes sufficiently small.

3.2 Finite-difference method

The finite-difference method uses the discretization derived in Section 2.5. The radial domain is sampled on a one-dimensional grid, and the differential equation is replaced by a linear algebraic system for the values of the potential at the half-grid points. The resulting matrix is tridiagonal, which makes the numerical solution efficient and straightforward.

This method is particularly useful for validating the discretization on simple benchmark cases. In particular, the uniform case provides an exact analytical solution against which the numerical approximation can be compared directly.

3.3 Finite-element method

The finite-element method is based on the variational formulation derived in Sections 2.7–2.9. The interval $[0, R]$ is divided into elements, and the potential is approximated by piecewise linear basis functions. The global linear system is assembled from the local stiffness matrices and local load vectors computed on each element.

This formulation is well suited to problems with piecewise material properties, since the dielectric coefficient $\varepsilon_r(r)$ can be evaluated element by element. It also provides a natural treatment of the regularity condition at the origin through the weak formulation.

4 Simulations and analyses

4.1 (a) Shooting Method with RK4

In this section, we solve the boundary value problem introduced in Section 2.1(d) using a shooting method combined with a fourth-order Runge–Kutta scheme. The objective is to determine the unknown value of the electric field at the outer boundary, $E_r(R)$, such that the regularity condition at the origin is satisfied.

As derived in Section 2.4(d), we use the following

$$y_1(x) = \phi(r), \quad y_2(x) = -E_r(r).$$

For this numerical test, we use the case studied analytically in Section 2.3(c):

$$\varepsilon_r(r) = 1, \quad \frac{\rho_{\text{lib}}(r)}{\varepsilon_0} = a_0 \sin\left(\frac{\pi r}{R}\right), \quad \phi(R) = 0.$$

Using the change of variable, the source term becomes

$$\sin\left(\frac{\pi r}{R}\right) = \sin(\pi(1-x)) = \sin(\pi x).$$

Thus, the system solved numerically is

$$\boxed{\frac{d}{dx} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} -Ry_2 \\ Ra_0 \sin(\pi x) + \frac{y_2}{1-x} \end{pmatrix}}. \quad (77)$$

The boundary condition at the outer radius is

$$\phi(R) = 0.$$

Since $r = R$ corresponds to $x = 0$, we impose

$$y_1(0) = 0.$$

The value of the electric field at $r = R$ is not known a priori. It is therefore used as the shooting parameter:

$$E_R = E_r(R).$$

Since $y_2 = -E_r$, the initial condition for a given shooting guess is

$$\boxed{y_1(0) = 0, \quad y_2(0) = -E_R}. \quad (78)$$

For each value of E_R , the system is integrated from $x = 0$ to $x = 1$ using a fourth-order Runge–Kutta method with 64 time steps. The correct value of E_R is the one that satisfies the inner boundary condition at the origin. By cylindrical symmetry,

$$E_r(0) = 0.$$

Since $y_2 = -E_r$, this condition becomes

$$\boxed{y_2(1) = 0.} \quad (79)$$

We therefore define the shooting residual as

$$F(E_R) = y_2(1; E_R),$$

and look for the root

$$F(E_R) = 0.$$

The root-finding procedure was performed using the secant method. Given two successive guesses $E_R^{(m-1)}$ and $E_R^{(m)}$, the next approximation is computed as

$$E_R^{(m+1)} = E_R^{(m)} - F(E_R^{(m)}) \frac{E_R^{(m)} - E_R^{(m-1)}}{F(E_R^{(m)}) - F(E_R^{(m-1)})}.$$

The analytical reference value obtained in Section 2.3(c) is

$$\boxed{E_r(R) = \frac{a_0 R}{\pi}.} \quad (80)$$

Using

$$a_0 = 10^4 \text{ V/m}^2, \quad R = 0.1 \text{ m},$$

we obtain

$$\boxed{E_r(R)_{\text{exact}} = \frac{10^4 \cdot 0.1}{\pi} \approx 318.310 \text{ V/m}.} \quad (81)$$

The shooting method with RK4 and 64 time steps gives

$$\boxed{E_r(R)_{\text{RK4}} = 318.306 \text{ V/m}.} \quad (82)$$

The corresponding absolute error is

$$|E_r(R)_{\text{RK4}} - E_r(R)_{\text{exact}}| = 3.99 \cdot 10^{-3} \text{ V/m}.$$

The relative error is

$$\frac{|E_r(R)_{\text{RK4}} - E_r(R)_{\text{exact}}|}{E_r(R)_{\text{exact}}} = 1.2 \cdot 10^{-5}.$$

The value of the final shooting residual was

$$F(E_R) = y_2(1) = 1.7 \cdot 10^{-14}.$$

This confirms that the shooting parameter has been adjusted correctly, since the residual is really close to zero and therefore the condition $E_r(0) = 0$ is satisfied. Figure 1 shows the electric field obtained with the RK4 shooting method and compares it with the analytical solution from Section 2.3(c). The numerical solution is expected to follow the analytical curve closely. In particular, the field should vanish at the origin and reach its maximum value at the outer boundary.

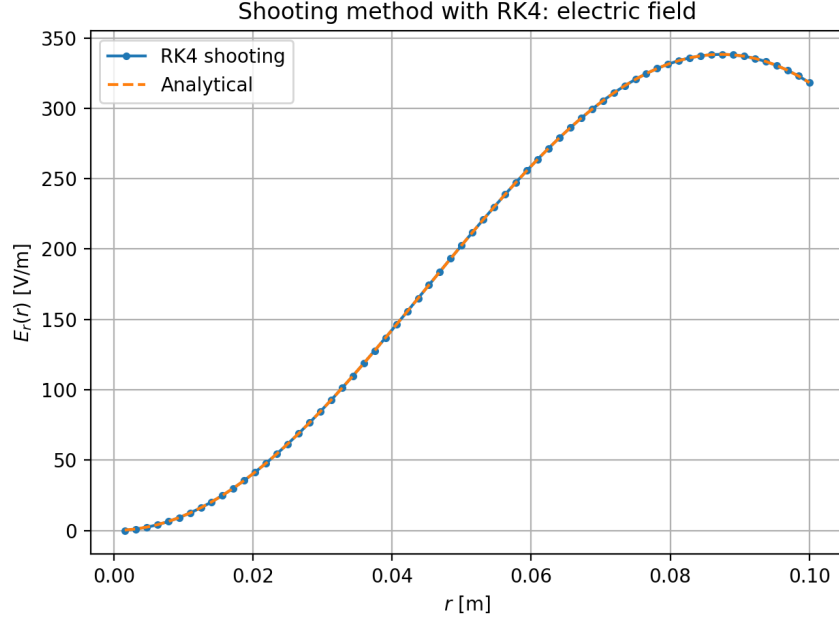


Figure 1: Electric field obtained with the RK4 shooting method using 64 time steps, compared with the analytical solution from Section 4.1(c).

Figure 2 shows the convergence of the secant method applied to the shooting residual. The residual decreases as the shooting parameter approaches the correct value of $E_r(R)$.

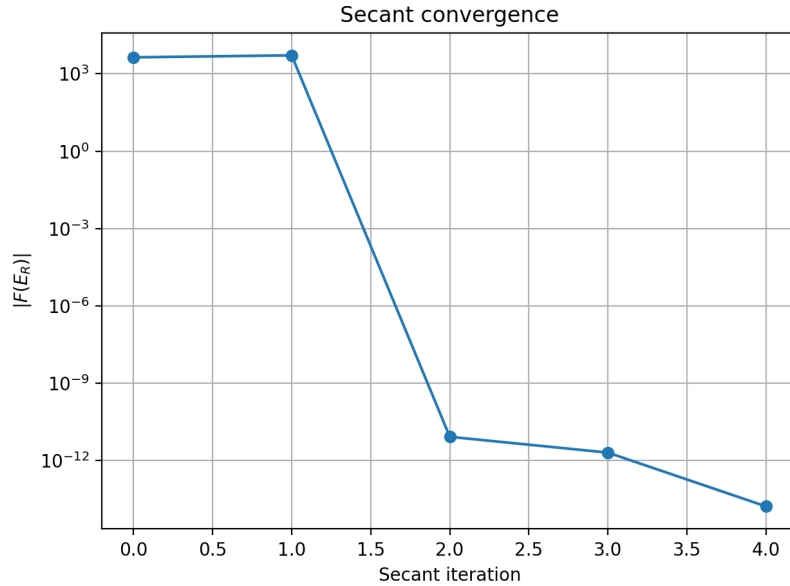


Figure 2: Convergence of the secant method for the residual $F(E_R) = y_2(1; E_R)$.

A numerical difficulty appears near $x = 1$, because the equation contains the term

$$\frac{y_2}{1-x}.$$

For the exact regular solution, this term remains finite because the imposed condition is $y_2(1) = 0$. Numerically, however, the endpoint must be handled carefully. In the implementation, the integration was stopped at the last grid point before $x = 1$, namely at $x = 1 - \Delta x$ with $\Delta x = 1/64$.

The shooting residual was evaluated at this point, which avoids the explicit division by zero while remaining close enough to the origin to impose the condition $E_r(0) = 0$ numerically.

Overall, the shooting method successfully transforms the boundary value problem into a family of initial value problems. By varying the unknown value $E_r(R)$ with the secant method, the internal symmetry condition $E_r(0) = 0$ is enforced. The comparison with the analytical result from Section 2.3(c) shows that the RK4 shooting method gives an accurate approximation of the correct outer electric field.

(Facultatif) Convergence study We performed a convergence study of the RK4 shooting integration. In this test, the shooting parameter was not obtained using the secant method. Instead, we used the exact initial value of the electric field at the outer boundary, derived in Section 2.3(c):

$$E_r(R) = \frac{a_0 R}{\pi}.$$

Therefore, the initial conditions were fixed as

$$y_1(0) = 0, \quad y_2(0) = -\frac{a_0 R}{\pi}.$$

The same first-order system as in the previous section was then integrated with RK4 for different values of the time step

$$\Delta x = \frac{1}{N}.$$

The numerical solution was compared with the analytical solution at two different radial positions. The first point was

$$r = \frac{R}{2},$$

which corresponds to

$$x = \frac{1}{2}.$$

The second point was the origin,

$$r = 0,$$

which corresponds to

$$x = 1.$$

For each value of Δx , we computed the errors

$$e_{R/2} = \left| E_{r,\text{num}}\left(\frac{R}{2}\right) - E_{r,\text{exact}}\left(\frac{R}{2}\right) \right|,$$

and

$$e_0 = |E_{r,\text{num}}(0) - E_{r,\text{exact}}(0)|.$$

Since the exact solution satisfies

$$E_{r,\text{exact}}(0) = 0,$$

the error at the origin reduces to

$$e_0 = |E_{r,\text{num}}(0)|.$$

Figure 3 shows the convergence curves obtained for both errors as a function of the time step Δx .

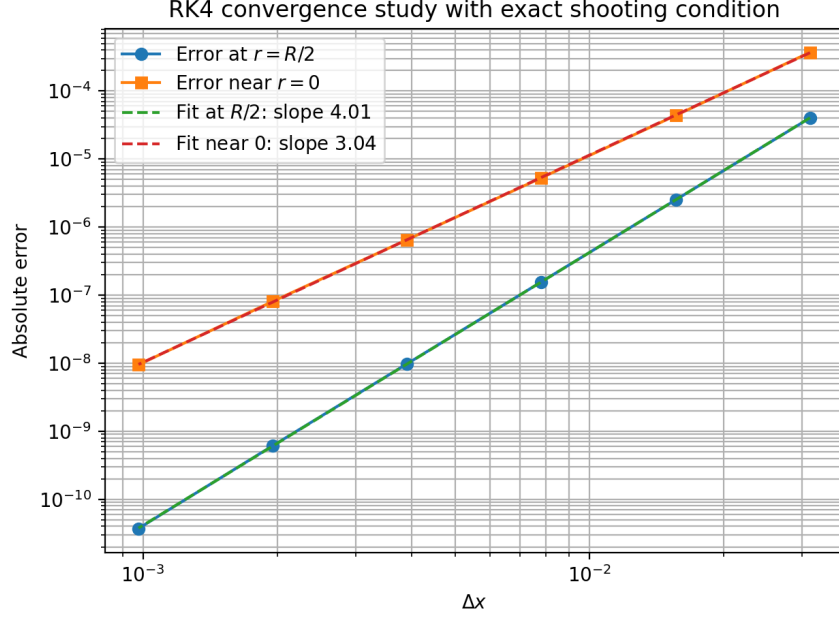


Figure 3: Convergence of the RK4 shooting integration using the exact initial value $E_r(R) = a_0 R/\pi$. The errors are measured at $r = R/2$ and near $r = 0$.

The estimated convergence order at $r = R/2$ was

$$p_{R/2} \approx 4.01.$$

The estimated convergence order near the origin was

$$p_0 \approx 3.04.$$

At the midpoint $r = R/2$, the error decreases approximately as expected for a fourth-order Runge–Kutta method:

$$e_{R/2} \sim \Delta x^4.$$

This is because the solution is smooth away from the origin, and the differential system does not present any numerical singularity at this point. Near the origin, however, the convergence is worse. This is due to the term

$$\frac{y_2}{1-x}$$

in the differential equation, which becomes singular as $x \rightarrow 1$. For the exact regular solution this term remains finite because $y_2(1) = 0$, but numerically small errors in y_2 are amplified near the endpoint. As a result, the error at $r = 0$ does not follow the ideal fourth-order RK4 convergence as cleanly as the error at $r = R/2$.

This does not indicate a failure of the RK4 method. Rather, it shows that the endpoint $r = 0$ is a numerically delicate point in cylindrical coordinates. Away from the origin, the expected fourth-order behaviour is recovered, while at the origin the apparent singularity of the equations reduces the observed convergence rate. The cylindrical coordinate singularity at the origin makes the endpoint more delicate, and the numerical error there is more sensitive to the treatment of the final integration step.

4.2 (b) Validation of the finite-element implementation

In this subsection, we validate the finite-element implementation by comparing the numerical solution with the analytical solution of the uniform test case derived in Section 2.2. In this

configuration, the dielectric constant is uniform,

$$\varepsilon_r(r) = 1,$$

the free charge density is constant,

$$\rho_{\text{lib}}(r) = \varepsilon_0,$$

and the boundary condition at the outer radius is

$$\phi(R) = 0.$$

For this case, the analytical potential and electric field are

$$\phi(r) = \frac{R^2 - r^2}{4}, \quad E_r(r) = \frac{r}{2}.$$

The purpose of this test is twofold. First, we check that the finite-element assembly and the treatment of the boundary condition produce the correct solution profiles. Second, we study how the numerical result converges as the mesh is refined.

Comparison of the numerical solution with the analytical one. We first perform a representative simulation with a fixed mesh and compare the finite-element solution with the analytical expressions. Figure 4 shows the radial electric field as a function of r , while Figure 5 shows the potential.

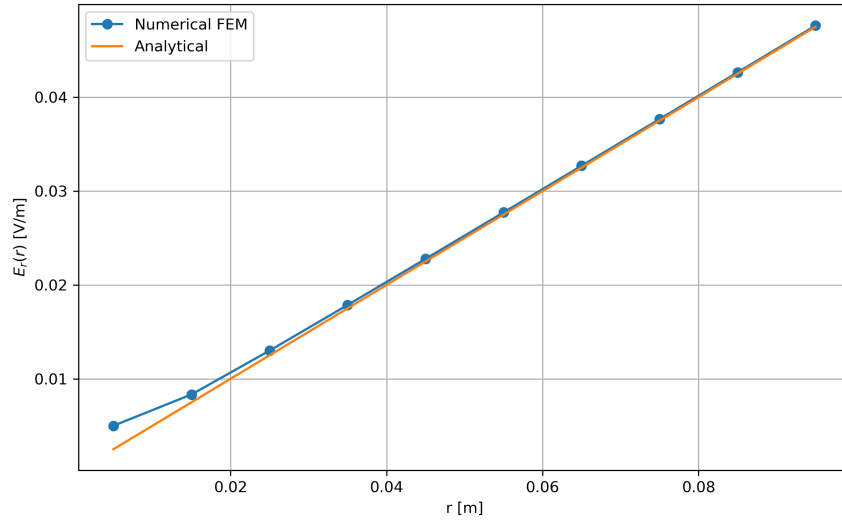


Figure 4: Radial electric field obtained with the finite-element method in the uniform test case, compared with the analytical solution $E_r(r) = r/2$.

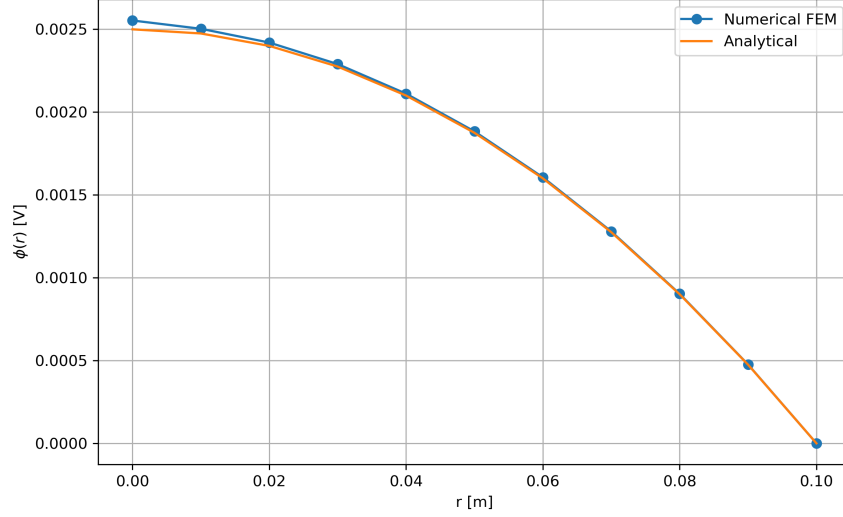


Figure 5: Electrostatic potential obtained with the finite-element method in the uniform test case, compared with the analytical solution $\phi(r) = (R^2 - r^2)/4$.

The agreement is very good in both figures. The electric field follows the expected linear dependence on r , and the potential reproduces the expected quadratic profile. Small discrepancies are only visible near the origin for the coarsest part of the discrete representation, but the numerical and analytical curves remain very close over the whole interval. This confirms that the implementation correctly reproduces the basic behaviour of the electrostatic solution in cylindrical geometry.

Convergence with mesh refinement. We then perform a convergence study by refining the mesh, taking

$$N_1 = N_2 = N,$$

and monitoring the numerical value of the potential at the origin, $\phi_h(0)$. In the present test case, the exact reference value is

$$\phi(0) = \frac{R^2}{4}.$$

Figure 6 shows the numerical value of $\phi_h(0)$ as a function of N , together with the exact value. As the mesh is refined, the numerical result approaches the analytical one.

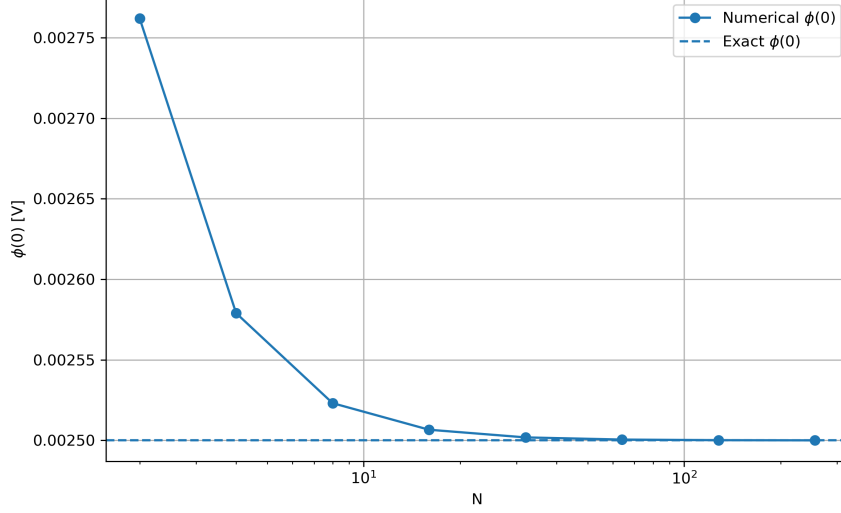


Figure 6: Numerical value of $\phi_h(0)$ as a function of the number of elements N , compared with the exact value $\phi(0) = R^2/4$.

To quantify this behaviour, we also plot the absolute error

$$|\phi_h(0) - \phi(0)|$$

as a function of N on logarithmic scales. The result is shown in Figure 7. The error decreases monotonically as the mesh is refined, which confirms the convergence of the finite-element approximation.

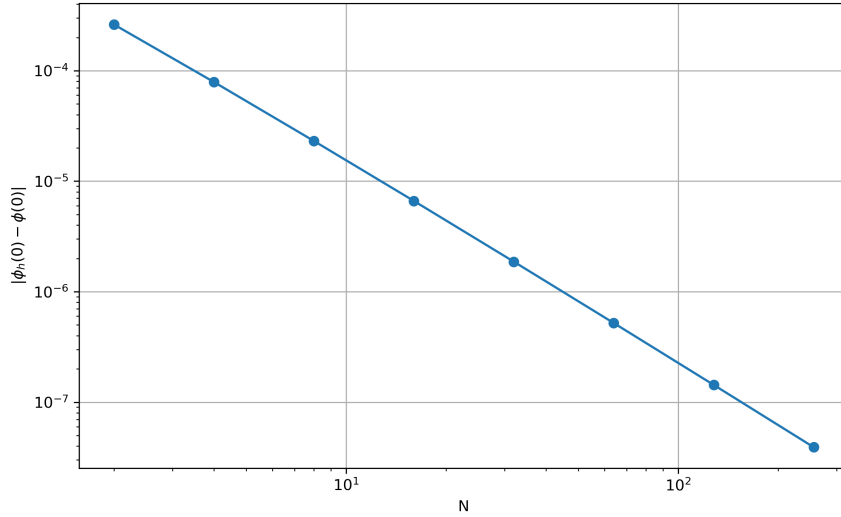


Figure 7: Absolute error on the potential at the origin as a function of the number of elements N for the finite-element method.

Overall, these results validate the finite-element implementation in the simplest benchmark configuration. The numerical profiles of $\phi(r)$ and $E_r(r)$ are consistent with the analytical solution, and the error on $\phi(0)$ decreases systematically when the mesh is refined. This shows that the assembly of the finite-element matrix, the treatment of the source term, and the implementation of the boundary condition are all correct.

4.3 (c) Non-trivial dielectric and free-charge distribution

We now consider the non-trivial configuration proposed in the exercise, with

$$b = 0.02 \text{ m}, \quad R = 0.1 \text{ m}, \quad V_0 = 0, \quad a_0 = 10^4 \text{ V/m}^2.$$

In this case, the dielectric profile and the free-charge density are no longer uniform. The relative permittivity is given by

$$\varepsilon_r(r) = \begin{cases} 1, & 0 \leq r < b, \\ 3 + 6 \frac{r-b}{R-b}, & b \leq r \leq R, \end{cases}$$

while the free-charge density is

$$\frac{\rho_{\text{lib}}(r)}{\varepsilon_0} = \begin{cases} a_0 \sin\left(\frac{\pi r}{b}\right), & 0 \leq r \leq b, \\ 0, & b < r \leq R. \end{cases}$$

The problem was solved with the finite-element method for several mesh refinements. In order to keep a comparable spatial resolution in the two subregions, we chose

$$N_2 = 4N_1,$$

which is consistent with the fact that the outer interval has length $R - b = 0.08 \text{ m}$, whereas the inner interval has length $b = 0.02 \text{ m}$. The simulations were carried out for

$$(N_1, N_2) = (5, 20), (10, 40), (20, 80), (40, 160).$$

Potential profile. Figure 8 shows the numerical potential $\phi(r)$ obtained for the different meshes. The four curves are almost indistinguishable on the scale of the plot, which indicates that the finite-element approximation of the potential is already very well converged, even for the coarsest mesh considered. This means that the global solution for $\phi(r)$ is numerically very stable with respect to mesh refinement.

From a physical point of view, the potential remains continuous over the whole domain, as expected in electrostatics, and decreases monotonically from the cylinder axis toward the outer boundary, where the imposed condition

$$\phi(R) = 0$$

is satisfied. A change of slope is visible around $r = b$, reflecting the transition between the charged inner region and the outer region with variable dielectric coefficient.

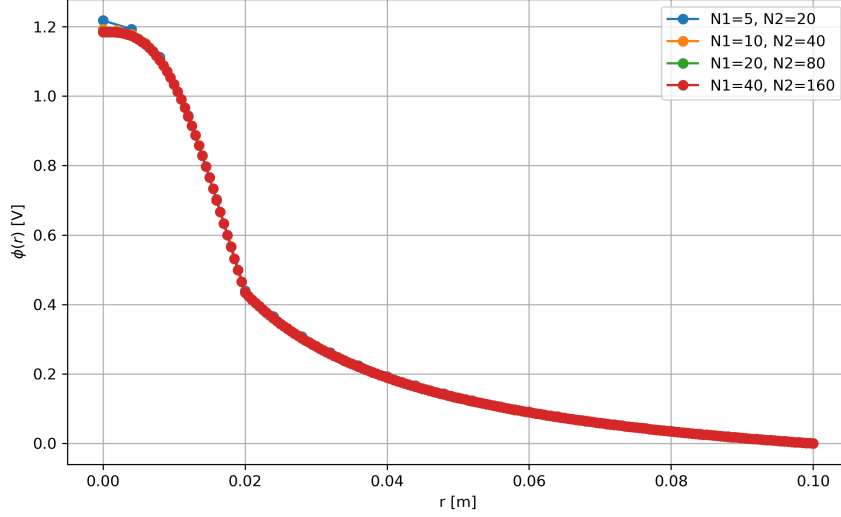


Figure 8: Potential $\phi(r)$ obtained with the finite-element method for the non-trivial case and for several mesh refinements.

Electric field. The corresponding electric field $E_r(r)$ is shown in Figure 9. Here again, the four numerical curves remain globally close to one another, which confirms the convergence of the finite-element solution as the mesh is refined. In this case, the effect of mesh refinement is most visible near the interface $r = b$, where the dielectric coefficient changes abruptly. Small differences between the meshes are also visible near the origin, whereas in the rest of the domain the curves tend to collapse progressively as the mesh is refined.

Inside the region $0 \leq r \leq b$, the field increases from zero and reaches a maximum near the interface. At $r = b$, a clear jump is observed. This behaviour is physically consistent: the normal component of the displacement field remains continuous at the interface in the absence of a free surface charge, whereas the electric field satisfies

$$E_r = \frac{D_r}{\varepsilon_0 \varepsilon_r}.$$

Therefore, a discontinuity in ε_r produces a discontinuity in E_r . Since ε_r jumps from 1 to 3 at $r = b$, the electric field decreases sharply at that point. For $r > b$, the field then decreases smoothly toward the outer boundary.

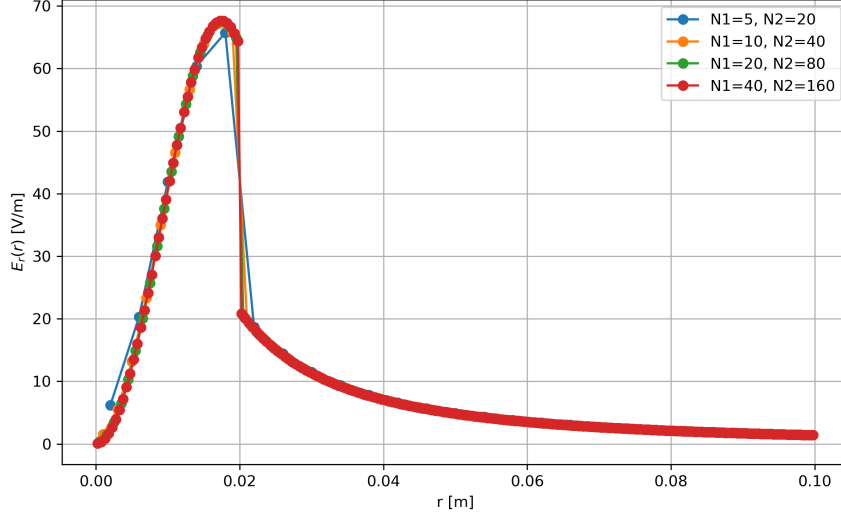


Figure 9: Radial electric field $E_r(r)$ obtained with the finite-element method for the non-trivial case and for several mesh refinements.

Displacement field. Figure 10 shows the quantity $D_r(r)/\varepsilon_0$. The curves corresponding to the different meshes show a clear convergence trend, although the differences between resolutions remain more visible than in the case of the potential. In particular, the coarsest mesh still departs slightly from the finer ones, whereas the solutions obtained with the refined meshes become much closer to each other. This indicates that the displacement field is also well converged numerically, but that its convergence is visually less immediate than for $\phi(r)$.

In the inner region, where free charge is present, the displacement field increases from the origin up to a maximum close to $r = b$. Beyond the interface, the free-charge density vanishes, so the equation

$$\nabla \cdot \mathbf{D} = \rho_{\text{lib}}$$

reduces to

$$\frac{1}{r} \frac{d}{dr} (r D_r) = 0,$$

which implies

$$D_r \propto \frac{1}{r}.$$

This is consistent with the smooth decay observed in the outer region. In contrast with the electric field, D_r remains continuous across the interface, which is again the expected physical behaviour.

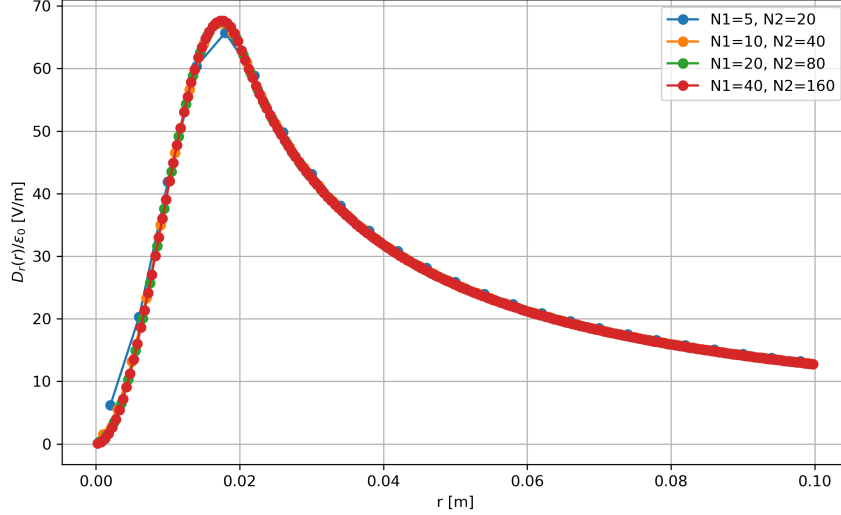


Figure 10: Displacement field $D_r(r)/\varepsilon_0$ obtained with the finite-element method for the non-trivial case and for several mesh refinements.

Discussion. The three figures together provide both a numerical and a physical validation of the solution. From the numerical point of view, the curves obtained for the four meshes are already very close to each other, showing that the finite-element approximation is well converged over the whole domain. The mesh dependence is very weak for the potential, somewhat more visible for the displacement field, and most noticeable near the interface and near the origin in the electric field.

From the physical point of view, the solution displays all the expected properties of the problem: the potential is continuous, the electric field exhibits a jump at the interface because of the discontinuity in the dielectric coefficient, and the displacement field remains continuous across the interface and decays approximately as $1/r$ in the outer charge-free region. Overall, these results confirm that the finite-element implementation correctly handles both the piecewise charge distribution and the spatially varying dielectric profile.

4.4 (d) Numerical verification of $\nabla \cdot \mathbf{D} = \rho_{\text{lib}}$

In this subsection, we verify numerically that the finite-element solution obtained in part (c) satisfies the electrostatic equation

$$\nabla \cdot \mathbf{D} = \rho_{\text{lib}}.$$

More precisely, after computing the displacement field $D_r(r)$, we evaluate its divergence numerically by finite differences and compare the result with the prescribed free-charge density. As in the previous subsection, the comparison is performed in normalized form, namely

$$\frac{\nabla \cdot \mathbf{D}}{\varepsilon_0} \quad \text{and} \quad \frac{\rho_{\text{lib}}}{\varepsilon_0}.$$

The same non-trivial configuration as in part (c) is used:

$$b = 0.02 \text{ m}, \quad R = 0.1 \text{ m}, \quad V_0 = 0, \quad a_0 = 10^4 \text{ V/m}^2,$$

with

$$\varepsilon_r(r) = \begin{cases} 1, & 0 \leq r < b, \\ 3 + 6 \frac{r-b}{R-b}, & b \leq r \leq R, \end{cases}$$

and

$$\frac{\rho_{\text{lib}}(r)}{\varepsilon_0} = \begin{cases} a_0 \sin\left(\frac{\pi r}{b}\right), & 0 \leq r \leq b, \\ 0, & b < r \leq R. \end{cases}$$

The verification was carried out for the same sequence of meshes as before,

$$(N_1, N_2) = (5, 20), (10, 40), (20, 80), (40, 160),$$

with $N_2 = 4N_1$ in order to keep comparable spatial resolution in the two subregions.

Comparison between $(\nabla \cdot \mathbf{D})/\varepsilon_0$ and $\rho_{\text{lib}}/\varepsilon_0$. Figure 11 shows the quantity $(\nabla \cdot \mathbf{D})/\varepsilon_0$ reconstructed numerically from the displacement field. As expected, it is non-zero only in the inner region $0 \leq r \leq b$, where free charge is present, and it tends to zero in the outer region $r > b$, where $\rho_{\text{lib}} = 0$. The numerical curves reproduce the expected bell-shaped profile and become more regular as the mesh is refined.

Figure 12 shows the imposed free-charge density $\rho_{\text{lib}}/\varepsilon_0$ on the same meshes. Its shape corresponds to the prescribed sinusoidal profile in the inner region and vanishes identically outside. Comparing Figures 11 and 12, we observe that the numerical divergence of $\mathbf{D}$ approaches the prescribed charge density as the mesh is refined.

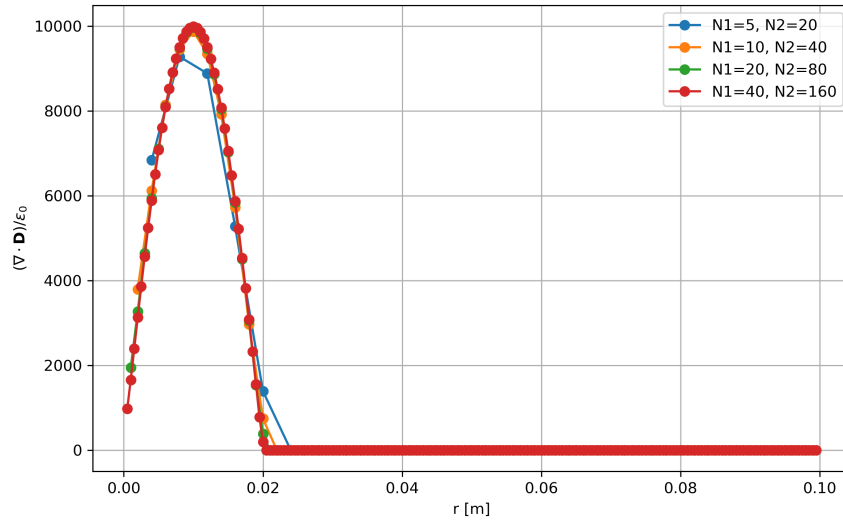


Figure 11: Numerical reconstruction of $(\nabla \cdot \mathbf{D})/\varepsilon_0$ for the non-trivial case and for several mesh refinements.

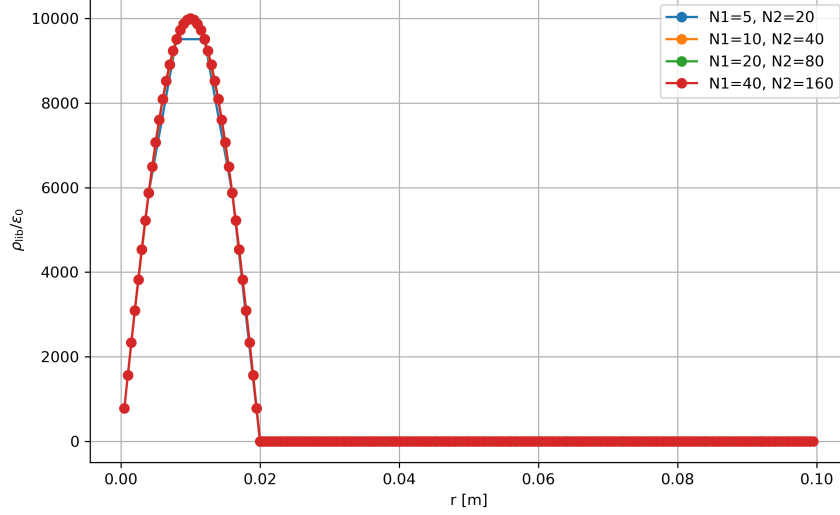


Figure 12: Prescribed free-charge density $\rho_{\text{lib}}/\varepsilon_0$ for the non-trivial case and for several mesh refinements.

Residual of the verification. The most direct way to assess the quality of the verification is to examine the residual

$$\frac{\nabla \cdot \mathbf{D} - \rho_{\text{lib}}}{\varepsilon_0},$$

shown in Figure 13. If the numerical solution satisfied Gauss' law exactly at every point, this quantity would vanish identically. In practice, the residual is not zero because $\nabla \cdot \mathbf{D}$ is reconstructed numerically from discrete values of D_r , so a finite discretization error is unavoidable.

The figure shows that the residual decreases strongly as the mesh is refined. For the coarsest mesh, sizeable deviations are visible in the charged region and especially near the interface $r = b$. As the number of elements increases, the residual becomes much smaller and tends to zero over most of the domain. In the outer region $r > b$, the residual is already essentially zero for all meshes, which is consistent with the fact that both $\nabla \cdot \mathbf{D}$ and ρ_{lib} vanish there.

The largest residuals are observed near the origin and near the interface $r = b$. This is physically and numerically reasonable. The origin is a delicate point because of the cylindrical geometry, while the interface $r = b$ is the location where the charge density abruptly drops to zero and where the material properties change regime. These are precisely the regions where a finite-difference reconstruction of the divergence is expected to be least accurate on coarse meshes.

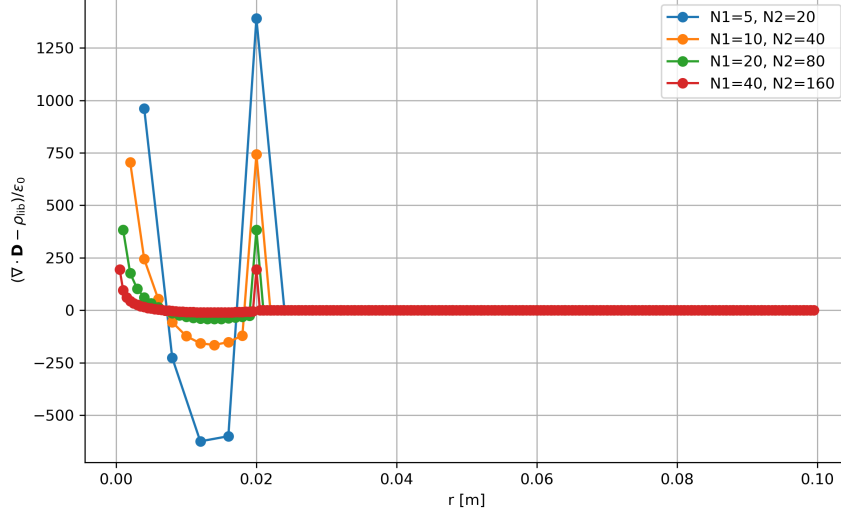


Figure 13: Residual $(\nabla \cdot \mathbf{D} - \rho_{\text{lib}})/\varepsilon_0$ for the non-trivial case and for several mesh refinements.

Discussion. These results provide a direct numerical validation of the electrostatic relation

$$\nabla \cdot \mathbf{D} = \rho_{\text{lib}}.$$

The reconstructed divergence of the displacement field reproduces the prescribed free-charge density with increasing accuracy as the mesh is refined. The residual tends toward zero throughout the domain, with the largest discrepancies confined to the most delicate regions, namely the origin and the interface at $r = b$. Overall, this confirms that the finite-element solution obtained in part (c) is not only qualitatively correct, but also quantitatively consistent with the fundamental local equation of the problem.

4.5 Modified $\varepsilon_r(r)$ and $\rho_{\text{lib}}(r)$

In this optional part, we explored a modified non-trivial configuration by changing both the dielectric profile and the free-charge distribution, while keeping the same finite-element implementation as in Section 4.3. The objective is to study how the numerical solution is affected by different physical choices for $\varepsilon_r(r)$ and $\rho_{\text{lib}}(r)$.

We considered the same geometric parameters as before,

$$b = 0.02 \text{ m}, \quad R = 0.1 \text{ m}, \quad V_0 = 0, \quad a_0 = 10^4 \text{ V/m}^2,$$

but replaced the previous non-trivial profiles by

$$\varepsilon_r(r) = \begin{cases} 2, & 0 \leq r < b, \\ 2 + 4 \frac{r-b}{R-b}, & b \leq r \leq R, \end{cases}$$

and

$$\frac{\rho_{\text{lib}}(r)}{\varepsilon_0} = \begin{cases} \frac{a_0}{2} \left(1 - \cos \left(\frac{2\pi r}{b} \right) \right), & 0 \leq r < b, \\ 0, & b \leq r \leq R. \end{cases}$$

The problem was solved for the same sequence of meshes as in the previous section,

$$(N_1, N_2) = (5, 20), (10, 40), (20, 80), (40, 160),$$

with $N_2 = 4N_1$ in order to keep a comparable spatial resolution in the two subregions.

Potential. The numerical potential $\phi(r)$ is shown in Figure 14. The four curves are almost superimposed, which indicates that the finite-element solution is already very well converged, even for moderately coarse meshes. The potential is continuous over the whole domain and decreases monotonically from the center of the cylinder toward the outer boundary, where the imposed Dirichlet condition

$$\phi(R) = 0$$

is satisfied.

Compared with the previous non-trivial case, the shape of the potential is now smoother around the interface $r = b$. This is consistent with the fact that the new dielectric profile is continuous at $r = b$, unlike the case studied in Section 4.3(c).

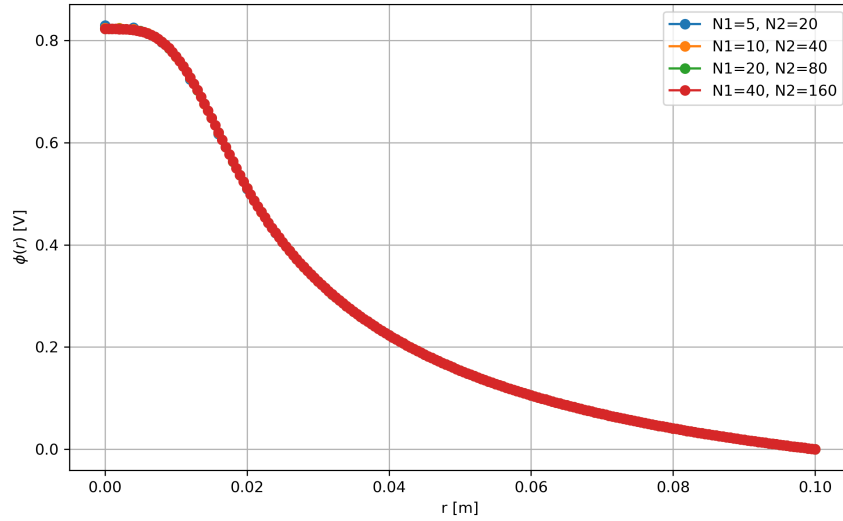


Figure 14: Potential $\phi(r)$ for the modified dielectric and free-charge profiles, for several mesh refinements.

Electric field. The radial electric field $E_r(r)$ is shown in Figure 15. It starts from zero at the origin, increases inside the charged region, reaches a maximum before the interface, and then decreases smoothly in the outer region. The numerical curves again show a clear convergence trend, with the finest meshes becoming almost indistinguishable.

A notable difference with respect to Section 4.3(c) is that no jump is observed at $r = b$. This is physically consistent, since the new dielectric profile satisfies

$$\varepsilon_r(b^-) = \varepsilon_r(b^+) = 2,$$

so the permittivity is continuous at the interface. As a result, the electric field also remains continuous there.

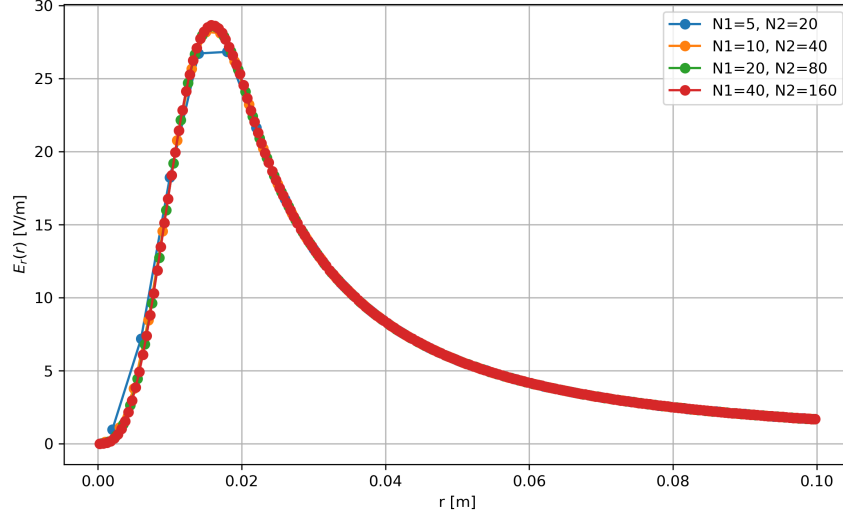


Figure 15: Radial electric field $E_r(r)$ for the modified dielectric and free-charge profiles, for several mesh refinements.

Displacement field. Figure 16 shows the corresponding displacement field $D_r(r)/\varepsilon_0$. As expected, it vanishes at the origin, increases in the inner charged region, and then decreases in the outer region where no free charge is present. The convergence with mesh refinement is again very good, the coarsest mesh showing only small deviations near the origin.

The smooth behaviour of D_r is also consistent with the modified charge density, which now rises and falls continuously inside the region $0 \leq r \leq b$. In the outer region, where $\rho_{\text{lib}} = 0$, the displacement field decreases gradually as expected from Gauss' law in a charge-free cylindrical region.

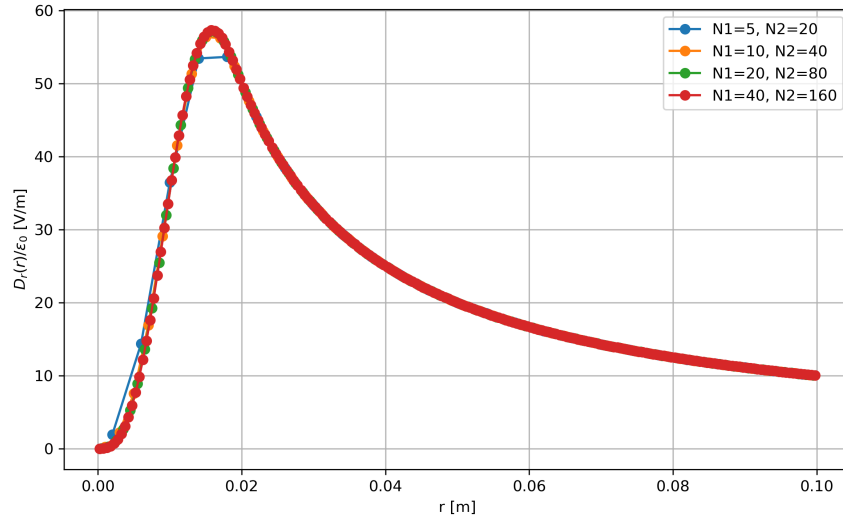


Figure 16: Displacement field $D_r(r)/\varepsilon_0$ for the modified dielectric and free-charge profiles, for several mesh refinements.

Discussion. This optional extension shows that the finite-element method remains robust when the physical profiles are modified. The numerical solutions for $\phi(r)$, $E_r(r)$, and $D_r(r)$ converge well as the mesh is refined, and the resulting profiles are physically consistent with the new choices of dielectric coefficient and free-charge density.

The main qualitative difference with the previous non-trivial case is the disappearance of the jump in the electric field at $r = b$. This is directly explained by the continuity of $\varepsilon_r(r)$ at the interface in the modified model. Overall, these results illustrate how the finite-element solution adapts naturally to different physical configurations while preserving numerical stability and convergence.

5 Conclusion

In this exercise, we derived and solved an electrostatic problem in cylindrical symmetry with free charges and variable dielectric properties. The analytical solutions obtained in the simple cases provided useful references for validating the numerical methods.

The RK4 shooting method, combined with a secant root-finding procedure, accurately recovered the exact value of $E_r(R)$. The convergence study showed fourth-order behaviour away from the origin, while the convergence near $r = 0$ was slightly degraded by the apparent singularity of the cylindrical formulation.

The finite-element method was successfully validated in the uniform case and then applied to the non-trivial dielectric and charge profiles. The computed potential, electric field and displacement field showed the expected physical behaviour and converged under mesh refinement. The numerical verification of $\nabla \cdot \mathbf{D} = \rho_{\text{lib}}$ further confirmed the consistency of the solution.

Overall, the simulations demonstrate that the implemented methods are accurate and robust for this class of cylindrically symmetric electrostatic problems.

6 References

1. Exercise statement: *Physique Numérique – Exercice 4, Electrostatique : cylindre avec diélectrique variable et charges libres. Méthode de tir, différences finies, éléments finis..*
2. L. Villard, *Physique Numérique*.
3. Use of LLM tools (ChatGPT) for language correction and report rephrasing.